What's new

On several irrationality problems for Ahmes series

Terence Tao · 27 November 2024

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Vjeko Kovac and I have just uploaded to the arXiv our paper "On several irrationality problems for Ahmes series". This paper resolves (or at least makes partial progress on) some open questions of Erdős and others on the irrationality of *Ahmes series*, which are infinite series of the form $\sum_{k=1}^\infty \frac{1}{a_k}$ for some increasing sequence a_k of natural numbers. Of course, since most real numbers are irrational, one expects such series to "generically" be irrational, and we make this intuition precise (in both a probabilistic sense and a Baire category sense) in our paper. However, it is often difficult to establish the irrationality of any specific series. For example, it is already a non-trivial result of Erdős that the series $\sum_{k=1}^\infty \frac{1}{2^k-1}$ is irrational, while the irrationality of $\sum_{p \hbox{ prime}} \frac{1}{2^p-1}$ (equivalent to Erdős problem #69) remains open, although very recently Pratt established this conditionally on the Hardy–Littlewood prime tuples conjecture. Finally, the irrationality of $\sum_n \frac{1}{n!-1}$ (Erdős problem #68) is completely open.

On the other hand, it has long been known that if the sequence a_k grows faster than C^{2^k} for any C, then the Ahmes series is necessarily irrational, basically because the fractional parts of $a_1 \dots a_m \sum_{k=1}^\infty \frac{1}{a_k}$ can be arbitrarily small positive quantities, which is inconsistent with $\sum_{k=1}^\infty \frac{1}{a_k}$ being rational. This growth rate is sharp, as can be seen by iterating the identity $\frac{1}{n} = \frac{1}{n+1} + \frac{1}{n(n+1)}$ to obtain a rational Ahmes series of growth rate $(C+o(1))^{2^k}$ for any fixed $C > 1$.

In our paper we show that if a_k grows somewhat slower than the above sequences in the sense that $a_{k+1} = o(a_k^2)$, for instance if $a_k \asymp 2^{(2-\varepsilon)^k}$ for a fixed $0 < \varepsilon < 1$, then one can find a comparable sequence $b_k \asymp a_k$ for which $\sum_{k=1}^\infty \frac{1}{b_k}$ is rational. This partially addresses Erdős problem #263, which asked if the sequence $a_k = 2^{2^k}$ had this property, and whether any sequence of exponential or slower growth (but with $\sum_{k=1}^\infty 1/a_k$ convergent) had this property. Unfortunately we barely miss a full solution of both parts of the problem, since the condition $a_{k+1} = o(a_k^2)$ we need just fails to cover the case $a_k = 2^{2^k}$, and also does not quite hold for all sequences going to infinity at an exponential or slower rate.

We also show the following variant; if a_k has exponential growth in the sense that $a_{k+1} = O(a_k)$ with $\sum_{k=1}^\infty \frac{1}{a_k}$ convergent, then there exists nearby natural numbers $b_k = a_k + O(1)$ such that $\sum_{k=1}^\infty \frac{1}{b_k}$ is rational. This answers the first part of Erdős problem #264 which asked about the case $a_k = 2^k$, although the second part (which asks about $a_k = k!$) is slightly out of reach of our methods. Indeed, we show that the exponential growth hypothesis is best possible in the sense a *random* sequence a_k that grows faster than exponentially will *not* have this property, this result does not address any specific superexponential sequence such as $a_k = k!$, although it does apply to some sequence a_k of the shape $a_k = k! + O(\log\log k)$.

Our methods can also handle higher dimensional variants in which multiple series are simultaneously set to be rational. Perhaps the most striking result is this: we can find an increasing sequence a_k of natural numbers with the property that $\sum_{k=1}^\infty \frac{1}{a_k+t}$ is rational for *every* rational t (excluding the cases $t = -a_k$ to avoid division by zero)! This answers (in the negative) a question of Stolarsky Erdős problem #266, and also reproves Erdős problem #265 (and in the latter case one can even make a_k grow double exponentially fast).

Our methods are elementary and avoid any number-theoretic considerations, relying primarily on the countable dense nature of the rationals and an iterative approximation technique. The first observation is that the task of representing a given number q as an Ahmes series $\sum_{k=1}^\infty \frac{1}{a_k}$ with each a_k lying in

some interval I_k (with the I_k disjoint, and going to infinity fast enough to ensure convergence of the series), is possible if and only if the infinite sumset

$$\frac{1}{I_1} + \frac{1}{I_2} + \dots$$

to contain q, where $\frac{1}{I_k} = \{\frac{1}{a} : a \in I_k\}$. More generally, to represent a tuple of numbers $(q_t)_{t\in T}$ indexed by some set T of numbers simultaneously as $\sum_{k=1}^\infty \frac{1}{a_k+t}$ with $a_k \in I_k$, this is the same as asking for the infinite sumset

$$E_1 + E_2 + \dots$$

to contain $(q_t)_{t\in T}$, where now

$$E_k = \{(\frac{1}{a+t})_{t\in T} : a \in I_k\}. \tag{1}$$

So the main problem is to get control on such infinite sumsets. Here we use a very simple observation:

> **Proposition 1 (Iterative approximation)** Let V be a Banach space, let $E_1, E_2, \dots$ be sets with each E_k contained in the ball of radius $\varepsilon_k > 0$ around the origin for some ε_k with $\sum_{k=1}^\infty \varepsilon_k$ convergent, so that the infinite sumset $E_1 + E_2 + \dots$ is well-defined. Suppose that one has some convergent series $\sum_{k=1}^\infty v_k$ in V, and sets $B_1, B_2, \dots$ converging in norm to zero, such that
>
> $$v_k + B_k \subset E_k + B_{k+1} \tag{2}$$
>
> for all $k \geq 1$. Then the infinite sumset $E_1 + E_2 + \dots$ contains $\sum_{k=1}^\infty v_k + B_1$.

Informally, the condition (2) asserts that E_k occupies all of $v_k + B_k$ "at the scale B_{k+1}".

Proof: Let $w_1 \in B_1$. Our task is to express $\sum_{k=1}^\infty v_k + w_1$ as a series $\sum_{k=1}^\infty e_k$ with $e_k \in E_k$. From (2) we may write

$$\sum_{k=1}^\infty v_k + w_1 = \sum_{k=2}^\infty v_k + e_1 + w_2$$

for some $e_1 \in E_1$ and $w_2 \in B_2$. Iterating this, we may find $e_k \in E_k$ and $w_k \in B_k$ such that

$$\sum_{k=1}^\infty v_k + w_1 = \sum_{k=m+1}^\infty v_k + e_1 + e_2 + \dots + e_m + w_{m+1}$$

for all m. Sending $m \to \infty$, we obtain

$$\sum_{k=1}^\infty v_k + w_1 = e_1 + e_2 + \dots$$

as required. □

In one dimension, sets of the form $\frac{1}{I_k}$ are dense enough that the condition (2) can be satisfied in a large number of situations, leading to most of our one-dimensional results. In higher dimension, the sets E_k lie on curves in a high-dimensional space, and so do not directly obey usable inclusions of the form (2); however, for suitable choices of intervals I_k, one can take some finite sums $E_{k+1} + \dots + E_{k+d}$ which will become dense enough to obtain usable inclusions of the form (2) once d reaches the dimension of the ambient space, basically thanks to the inverse function theorem (and the non-vanishing curvatures of the curve in question). For the Stolarsky problem, which is an infinite-dimensional problem, it turns out that one can modify this approach by letting d grow slowly to infinity with k.

Comments (6)

 **allenknutson** · 28 November 2024

You're misspelling Erdős (wrong umlaut).
[Corrected, thanks – T.]

 **woutervandoorn** · 28 November 2024

Small typo on page 9: it says 'by one the present authors'
[Thanks, this will be corrected in the next revision of the ms. -T]

 **Anonymous** · 29 November 2024

"**Proposition 1 (Iterative approximatiom)**" typo. Should be **Proposition 1 (Iterative approximation)" .**
[Corrected, thanks -T.]

 **Anonymous** · 1 December 2024

An amazing theorem!
Also,"althuogh" in the fourth paragraph should be "although".
[Corrected, thanks -T.]

 **Anonymous** · 13 December 2024

Very interesting read. Does this suggest any new "universal" sequences that can represent rational numbers in infinitely many ways?

 **Terence Tao** · 14 December 2024

Yes, our methods will allow one to construct various rapidly growing sequences of natural numbers A with the property that an infinite number of rational numbers can be represented in the form $\sum_{n\in B} 1/n$ for infinitely many subsets B of A. For instance, the explicit choice $A = \{2^n + b : n \geq 1, 1 \leq b \leq 5\}$ will work (all numbers in an open neighbourhood of 0.75, regardless of rationality will have infinitely many such representations, by a slight modification of the arguments already in the paper). On the other hand, if $a_n/2^n \to \infty$, then a simple union bound argument can show that the set of sums $\sum_{n\in B} 1/n$ for $B \subset \{a_1, a_2, ...\}$ has measure zero, and so it is highly unlikely that it can represent an infinite number of rational numbers (though this is not quite a rigorous argument).